Self-Supervised Temporal Phenotype Trajectory Analysis of ICU Sepsis Patients from Multi-Center Time-Series Data

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Abstract

Background. Sepsis exhibits substantial clinical heterogeneity, yet most phenotyping approaches rely on static summary features that compress temporal dynamics into single-timepoint representations. Whether learned temporal representations can reveal clinically meaningful phenotype trajectories within the first 48 ICU hours remains underexplored.

Methods. We developed a three-stage computational framework applied to 11,986 ICU patients from the PhysioNet 2012 multi-center database (4 hospitals, 2 centers). First, we established a static phenotyping baseline using statistical features and PCA. Second, we trained a Transformer-based encoder via self-supervised learning (masked value prediction combined with a temporal contrastive window objective) and compared four representation methods across clustering quality, mortality stratification, center stability, and missingness robustness. Third, using the selected self-supervised representation, we extracted rolling-window embeddings (24-hour windows, 6-hour stride) and performed descriptive temporal phenotype trajectory analysis via K-Means clustering on per-window embeddings. As supplementary downstream validation, we added four supervised analyses: a logistic-regression classifier on frozen S1.5 embeddings, a local bridge aligning 40,331 PhysioNet/CinC 2019 sepsis stays to 18 shared channels for auxiliary supervision, an end-to-end attention-pooled fine-tuning model, and a leakage-aware 5-fold out-of-fold (OOF) stacking committee built on development-split base predictions. We additionally characterized cohort composition and structured missingness to interpret why explicit mask-aware modeling is necessary, audited the stacked model with bootstrap confidence intervals, calibration analysis, and permutation importance over its meta-features, and performed a systematic calibration optimization via structural redesign of the meta-learner (removing class-weight inflation, reducing base learner complexity) combined with post-hoc Platt scaling and a 30-configuration hyperparameter search. We also executed the legacy V1 static pipeline end-to-end on full credentialed local MIMIC-IV 3.1 and eICU-CRD 2.0 extracts as supplementary external-database transfer analyses, and implemented a unified external temporal transfer runner that maps prepared MIMIC-IV / eICU cohorts into the 21-channel S0 schema, reuses PhysioNet 2012 preprocessing statistics, and executes the frozen S1.5 + Stage 3 workflow. Full-cohort external temporal runs were then executed on 94,458 MIMIC-IV ICU stays and 200,859 eICU stays. All mortality outcomes were obtained from verified PhysioNet Outcomes files (in-hospital mortality rate: 14.2%).

Results. Data analysis showed a steep missingness gradient between core hemodynamic variables (9.8%–11.9% missing for heart rate and blood pressure channels) and laboratory variables (mean 93.8% missing; bilirubin 98.3%). Static K=4 clustering on PCA-reduced features identified phenotypes with mortality rates ranging from 6.8% to 36.0%. The self-supervised encoder achieved superior center stability (cluster distribution L1 distance = 0.016 vs 0.027 for PCA) and reduced missingness sensitivity ($|r| = 0.15$ vs 0.23), while PCA retained stronger static mortality separation. A downstream mortality classifier trained on frozen S1.5 embeddings achieved AUROC 0.829 and balanced accuracy 0.745 on the held-out Center B test split. Auxiliary supervision on the bridged 2019 cohort followed by end-to-end fine-tuning raised held-out performance to accuracy 0.795, balanced accuracy 0.753, and AUROC 0.842. A 35-run accuracy-oriented downstream search further increased selected-model test accuracy to 0.871 and identified a validation-AUROC leader with test accuracy

0.874 and AUROC 0.867. A leakage-aware 5-fold OOF stacking committee then reached test accuracy 0.880 and AUROC 0.873; applying a balance-oriented threshold to the same probabilities yielded balanced accuracy 0.792, recall 77.6%, and F1 0.536. Bootstrap validation gave an AUROC 95% confidence interval of 0.858–0.888, but the initial stacked probabilities were poorly calibrated (Brier 0.144, ECE 0.222). A calibration-aware redesign—removing class-weight inflation from the meta-learner and reducing base learner complexity—resolved this deficit, achieving Brier 0.090, ECE 0.023, AUROC 0.873, and recall 83.8%, with mean predicted probability (14.1%) closely matching the observed mortality rate (14.6%). Temporal trajectory analysis revealed that 64.8% of patients maintained a stable phenotype across all windows, while 35.2% exhibited at least one transition. Stable phenotypes showed strongly stratified mortality (4.0%, 9.7%, 22.5%, 31.7%; range 27.7 percentage points). The three most frequent non-self transitions accounted for 48.1% of all transition events and predominantly moved toward lower-acuity phenotypes. Sensitivity analysis with reduced overlap (stride=12 h) confirmed identical mortality ordering and range (28.0 pp), with transition rates increasing under reduced overlap, indicating the primary analysis provides a conservative estimate. Cross-center temporal validation within the PhysioNet 2012 cohort (train on Center A, evaluate on Center B) confirmed identical mortality ordering, the same highest-risk phenotype, and a 25.8 percentage-point mortality range on the held-out center. Using the same frozen-transfer entry, supplementary external temporal runs processed 94,458 MIMIC-IV ICU stays and 200,859 eICU stays, yielding overall window silhouettes of 0.119 and 0.193, stable fractions of 50.9% and 43.1%, and non-self transition proportions of 14.3% and 17.3%, respectively. Legacy static transfer runs on the same full cohorts continued to show a dominant lower-risk cluster and a smaller shock-enriched higher-risk cluster.

Conclusions. Self-supervised temporal representations enable descriptive phenotype trajectory analysis that reveals clinically meaningful within-stay dynamics beyond static clustering. The same learned backbone also supports frozen, end-to-end, and stacked supervised mortality models, and auxiliary transfer from a bridged second ICU dataset can improve held-out discrimination. A calibration-aware redesign of the stacking meta-learner resolved the initial miscalibration (ECE 0.222→0.023, Brier 0.144→0.090) while preserving AUROC (0.873) and improving recall to 83.8%, producing probability estimates aligned with observed mortality rates and suitable for clinical risk communication. The data audit in this study shows that explicit treatment of missingness is not optional but foundational for ICU time-series modeling. Cross-center temporal validation within the PhysioNet 2012 multi-center cohort confirms that phenotype structure transfers across centers, though both centers derive from the same source database. Supplementary full-cohort frozen-transfer temporal runs on MIMIC-IV and eICU now broaden external evidence, while source-specific retraining and treatment-aware validation remain future work. These findings support further investigation with treatment-aware datasets, external database validation, and prospective study designs.

Keywords: sepsis phenotyping, self-supervised learning, temporal trajectories, ICU time-series, multi-center, representation learning

1 Introduction

1.1 Background

Sepsis affects approximately 49 million individuals worldwide annually and is associated with mortality rates ranging from 20% to over 50% in intensive care units (ICUs) [1]. Despite decades of clinical research, sepsis remains one of the most challenging conditions in critical care medicine. A key contributor to this challenge is the recognition that sepsis is not a single homogeneous disease but rather a syndrome encompassing multiple distinct biological phenotypes [2].

The landmark study by Seymour et al. (2019) identified four sepsis phenotypes—designated α, β, γ, and δ—with markedly different clinical trajectories, biomarker profiles, and treatment

responses [3]. More recently, Antcliffe et al. (2025) argued that the true value of subphenotyping lies in predicting heterogeneous treatment effects (theragnostics) rather than prognosis alone [14]. Feng et al. (2025) demonstrated that organ interaction trajectories can be modeled via deep temporal graph clustering with external validation [16].

1.2 Problem Statement

Existing approaches to sepsis subtyping face several challenges:

1. **Static representations**: Most methods compress 48-hour trajectories into summary statistics, discarding temporal dynamics that may distinguish improving from deteriorating patients.
2. **Data heterogeneity**: Different ICU databases use different variable nomenclature, measurement frequencies, and data formats.
3. **High missingness**: Clinical time-series data are inherently sparse, with laboratory values often measured only once or twice daily.
4. **Cross-center generalizability**: Subtypes discovered in a single center may not transfer to other hospitals.
5. **Outcome label quality**: Many studies rely on proxy mortality labels rather than verified discharge outcomes.

1.3 Contributions

This work makes the following contributions:

- A **three-stage phenotyping framework** progressing from static clustering to self-supervised representation learning to descriptive temporal phenotype trajectory analysis.
- A **self-supervised Transformer encoder** trained via masked value prediction and temporal contrastive learning, with systematic comparison against PCA baselines.
- **Descriptive temporal phenotype trajectories** showing that 35.2% of ICU sepsis patients undergo phenotype transitions within 48 hours, with transitions predominantly toward lower-acuity states.
- A **validated data pipeline** using ground-truth in-hospital mortality outcomes, replacing previously used proxy labels.
- A **cross-dataset bridge** aligning 40,331 PhysioNet/CinC 2019 sepsis stays to the S0 tensor interface, enabling auxiliary supervision with 18 shared physiologic channels.
- A **supplementary downstream validation suite** spanning frozen probes, end-to-end attention-pooled fine-tuning, systematic downstream hyperparameter search, and leakage-aware OOF stacking with bootstrap/calibration audits, clarifying why raw accuracy is misleading at 14% event prevalence.
- **Robustness validation** including stride sensitivity analysis and center probe analysis confirming no center information leakage.
- **Cross-center temporal validation** within the PhysioNet 2012 multi-center cohort, confirming that phenotype structure and mortality stratification transfer from the training center to a held-out center.

2 Related Work

2.1 Sepsis Phenotyping

The concept of sepsis heterogeneity was first systematically investigated by Seymour et al. (2019), who applied latent class analysis to 20,189 sepsis patients [3]. Their four-phenotype model has been replicated by subsequent studies [4, 5]. More recently, deep temporal clustering [6], contrastive learning [7], and organ interaction trajectory modeling [16] have been

proposed. Zhang et al. (2024) demonstrated time-series k-means with DTW for sepsis subphenotyping with external validation [17].

2.2 Self-Supervised Representation Learning for Clinical Data

Self-supervised representation learning (SSRL) has emerged as a powerful approach for extracting patient representations from unlabeled clinical data. Zheng et al. (2025) reviewed 46 SSRL studies, identifying Transformer-based (43%), autoencoder-based (28%), and graph-based (17%) model families [15]. Amirahmadi et al. (2025) introduced trajectory-ordered objectives that improve downstream prediction beyond standard masked modeling [18].

2.3 ICU Time-Series Analysis

ICU data present unique analytical challenges. The PhysioNet/CinC Challenge series has driven methodological advances [8, 9]. Huang et al. (2025) introduced MIMIC-Sepsis, a curated benchmark incorporating treatment variables that substantially improve model performance [19].

3 Methods

3.1 System Architecture

The system implements a three-stage framework (Figure 1):

```
Stage 1: Data -> Preprocessing -> Features -> PCA -> K-Means
Stage 2: Time Series + Masks -> Transformer -> Self-Supervised Pretraining ->
    Embeddings
Stage 3: Rolling Windows -> Per-Window Embedding -> Temporal Clustering -> Transitions
```

Each stage builds on the previous, with modules designed as independent components enabling systematic comparison across representation methods.

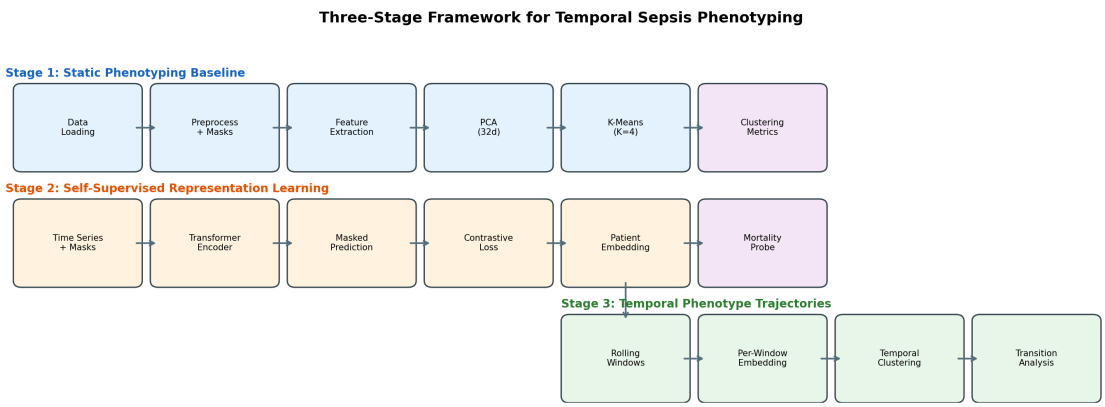

Figure 1: Three-stage framework for temporal sepsis phenotyping. Stage 1 establishes a static PCA baseline. Stage 2 trains a self-supervised Transformer encoder via masked prediction and contrastive learning. Stage 3 extracts rolling-window embeddings for descriptive temporal phenotype trajectory analysis.

3.2 Data Sources and Outcome Labels

3.2.1 PhysioNet 2012 Multi-parameter ICU Database

The primary dataset contains records from 12,000 ICU patients across four hospitals [8]. After quality filtering, 11,986 patients were retained:

- **Center A** (set-a + set-b): 7,989 patients
- **Center B** (set-c): 3,997 patients from a separate hospital within the PhysioNet 2012 cohort

Outcome labels. Ground-truth in-hospital mortality was obtained from PhysioNet Outcomes files for all 11,986 patients (mortality rate: 14.2%). An earlier version of this analysis used proxy mortality derived from physiological thresholds (GCS ≤ 5 or sustained MAP < 55 mmHg). Outcome audit revealed that this proxy had a positive predictive value of only 17.7%. All results in this paper use verified outcome labels.

Data schema. Following CLIF-inspired design principles [20], the data layer separates continuous measurements (21 variables), observation masks, and proxy-derived indicators into distinct tensors. Proxy indicators (MAP < 65, $FiO_2 > 0.21$) are explicitly labeled as proxies and stored separately from true treatment records, which are unavailable in PhysioNet 2012.

3.2.2 Supplementary Datasets

A configurable synthetic data generator with known ground-truth labels enables clustering recovery evaluation (ARI, NMI). In addition, we built a local bridge from the PhysioNet/CinC 2019 Sepsis Challenge stubs [9], yielding 40,331 ICU stays in the same on-disk format as `data/s0`. Eighteen of the 21 continuous PhysioNet 2012 channels map directly (`gcs`, `sodium`, and `pao2` are unavailable in the 2019 source), and PhysioNet 2012 preprocessing statistics are reused so the auxiliary source is numerically compatible with the pretrained encoder. During this revision we also unified external-database ingestion behind `src/main.py` and executed full credentialed local runs on MIMIC-IV 3.1 and eICU-CRD 2.0. The MIMIC run built a DuckDB database from raw CSVs, executed the concept SQL pipeline, exported 94,458 ICU stays including 41,295 Sepsis-3 stays (4,533,984 hourly rows), and then ran the legacy static clustering/evaluation path. The eICU run produced cached tensors for 200,859 ICU stays and the same downstream clustering/evaluation outputs. In parallel, we implemented `scripts/run_external_temporal_stage3.py`, which converts prepared MIMIC-IV / eICU cohorts into `data/external_temporal/<source>/s0`, reuses the PhysioNet 2012 preprocessing statistics, and then executes the frozen S1.5 + Stage 3 trajectory workflow. Full-cohort external temporal runs then completed successfully for both sources. The MIMIC temporal transfer bundle maps 15 of the 21 continuous S0 channels, leaving lactate, bilirubin, and blood gas channels structurally missing; the eICU temporal transfer bundle maps 12 of the 21 channels, leaving GCS, BUN, glucose, electrolytes, and blood gas channels missing. The static and temporal external runs are therefore best interpreted as supplementary frozen-transfer analyses under partial feature overlap rather than as source-specific retraining studies.

3.3 Preprocessing

All measurements are resampled to a uniform hourly grid over 48 hours using LOCF. Outliers exceeding 4σ are clipped. Missing values are imputed via forward fill (up to 6 hours), global median, and zero-fill as fallback. Observation masks recording which values were truly measured versus imputed are preserved as first-class inputs. Overall missing rate is 73.3%.

3.4 Static Feature Engineering (Stage 1)

From the preprocessed 3D tensor ($n \times 48 \times 21$), we extract a 2D feature matrix using six statistical functions (mean, std, min, max, trend, last value) across three windows (12 h, 24 h, 48 h), yielding 378 features plus derived clinical indicators. PCA reduces this to 32 dimensions (62.5% variance explained).

3.5 Self-Supervised Representation Learning (Stage 2)

3.5.1 Encoder Architecture

A Transformer-based encoder processes the hourly time series with observation masks as explicit input channels. The input is formed by concatenating value and mask vectors: $\mathbf{x}_{\text{input}} = [\mathbf{x}; \mathbf{m}] \in \mathbb{R}^{T \times 42}$. A linear projection maps this to $d_{\text{model}} = 128$ dimensions. Learnable positional encodings provide temporal position awareness. A 2-layer, 4-head Transformer encoder with pre-layer normalization and GELU activation produces per-timestep representations, aggregated via observation-density-weighted mean pooling.

3.5.2 Pretraining Objectives

Masked value prediction. 15% of observed values are randomly masked. The encoder must predict original values at masked positions. Loss: MSE on masked observed positions only.

Temporal contrastive window objective. Two stochastic 30-hour views are extracted with constrained overlap (12–24 hours). Both are encoded by the shared encoder, then projected through a 2-layer MLP projection head ($128 \to 128 \to 64$). NT-Xent loss ($\tau = 0.1$) with in-batch negatives. The projection head is discarded after pretraining.

Combined loss: $\mathcal{L} = \mathcal{L}_{\text{masked}} + \lambda(t) \cdot \mathcal{L}_{\text{contrastive}}$, where λ warms up linearly from 0 to 0.5 over 10 epochs.

3.5.3 Representation Comparison

Four representations are compared: (1) PCA baseline (32d), (2) masked reconstruction only (S1, 128d), (3) masked + contrastive (S1.5, λ=0.5, 128d), and (4) reduced contrastive ablation (S1.6, λ=0.2, 128d).

3.5.4 Principle Interpretation of the Representation Design

The Stage 2 design separates four complementary roles. First, concatenating values and observation masks preserves whether a measurement is truly observed or merely imputed, preventing the model from conflating absence with physiologic normality. Second, masked reconstruction forces the encoder to learn local temporal and cross-variable dependencies from real observations rather than memorizing complete trajectories. Third, the contrastive window objective aligns two partially overlapping 30-hour views from the same patient stay, encouraging patient-level consistency under local time shifts and sampling variation. Fourth, observation-density-weighted pooling gives more influence to hours carrying more measured signal without discarding sparse hours entirely. Together, these choices explain why the learned representation can be reused on arbitrary sub-windows in Stage 3 while remaining less sensitive to center mix and raw observation density than simpler baselines.

3.6 Descriptive Temporal Phenotype Trajectories (Stage 3)

Using the frozen S1.5 encoder, rolling-window embeddings are extracted: 5 windows of 24 hours with 6-hour stride. K-Means (K=4) is fit on train-split windows only (30,955 windows) and applied to all 59,930 windows. Patients are classified as *stable* (all 5 labels identical), *single-transition* (1 change), or *multi-transition* ($\geq$2 changes). Conceptually, each 24-hour embedding acts as a local clinical state summary, so Stage 3 converts patient stays into observable state sequences without imposing a parametric transition model. All temporal analyses are descriptive; no latent-state models or causal claims are involved.

3.7 Evaluation and Robustness

Clustering: Silhouette score, mortality separation. **Probes:** Mortality AUROC, center AUROC (↓ better), ICU LOS R^2, density-norm correlation. **Supplementary downstream validation:** In addition to a frozen logistic-regression probe on S1.5 embeddings, we trained an end-to-end attention-pooled classifier initialized from the S1.5 encoder and optionally warm-started with auxiliary supervision from the bridged PhysioNet 2019 cohort. We also ran a 35-candidate downstream hyperparameter search across LogisticRegression, HistGradientBoosting, and HGB ensemble variants over multiple feature views, followed by a 5-fold stratified OOF stacking committee trained on development-split base predictions from HGB and logistic branches. Because mortality prevalence is only ∼14%, we report AUROC, balanced accuracy, precision, recall, F1, and plain accuracy, compare accuracy against the majority-class baseline, and audit the final stacker with bootstrap confidence intervals, calibration statistics, and permutation importance over its meta-features. **Multi-seed:** 5 random seeds for all KMeans. **Stride sensitivity:** Repeated with stride=12 h (50% overlap). **Implementation:** Clustering and linear probes used scikit-learn [12]; figures were generated with Matplotlib [13].

4 Results

4.1 Cohort Characterization and Missingness Pattern

After quality filtering, 11,986 patients were retained (Center A: 7,989; Center B: 3,997). Ground-truth in-hospital mortality was 14.2%, and the retained cohort covered nearly the full 48-hour analytic window (mean ICU LOS 47.3 hours, median 47.5 hours).

Table 1: Cohort characterization and data completeness.

Metric	**Value**
Patients	11,986
Age	64.6 ± 17.2 years (median 67)
Sex	43.9% female, 56.0% male, 0.1% unknown
Center split	7,989 Center A / 3,997 Center B
In-hospital mortality	14.2% overall; 14.0% in Center A, 14.6% in Center B
ICU LOS	47.3 h mean; 47.5 h median
ICU type distribution	14.7% / 21.1% / 35.8% / 28.4% for types 1–4
Height / weight missing	47.7% / 8.3%
Core hemodynamic missingness	9.8%–11.9% for heart rate, SBP, DBP, MAP
Blood gas missingness	87.3% mean across PaO_2, FiO_2, $PaCO_2$, pH
Laboratory missingness	93.8% mean; bilirubin 98.3%, lactate 95.9%

The missingness pattern was strongly structured rather than random. Core hemodynamic channels were relatively dense, whereas respiratory, neurological, and laboratory variables were far sparser: respiratory rate was 75.9% missing, temperature 62.9%, GCS 67.9%, and most laboratory markers exceeded 92% missingness. This gradient explains why explicit observation masks are essential in this project: a model that sees only imputed values would incorrectly treat "not measured" as weak evidence of physiologic normality.

4.2 Static Phenotyping Baseline (Stage 1)

Static K=4 clustering on PCA-reduced statistical features identified phenotypes with mortality rates of 6.8%, 10.0%, 33.8%, and 36.0% (range 29.2 percentage points). Silhouette score was 0.061, consistent with the fuzzy but clinically stratified cluster boundaries expected in real ICU

data with severe sparsity. This baseline established that clinically meaningful risk separation was present in the cohort before any self-supervised learning was introduced.

4.3 Representation Learning Comparison (Stage 2)

Table 2: Representation comparison (K=4, mean ± std over 5 seeds).

Method	**Silhouette**	**Mort. Range**	**Center L1** ↓	**Mort. Probe**	$\lvert r\rvert_{\textbf{density}}$ ↓
PCA (32d)	0.061	29.2%	0.027	0.825	0.231
S1: masked (128d)	0.087	17.6%	0.024	0.825	0.247
S1.5: mask+contr. (128d)	0.080	24.6%	**0.016**	**0.830**	**0.148**
S1.6: λ=0.2 (128d)	0.079	25.1%	0.021	0.825	0.148

Center probe AUROC was 0.50–0.52 for all methods, confirming that no representation leaked center identity in a directly separable way. Compared with S1, adding the contrastive window objective in S1.5 slightly reduced geometric compactness (silhouette 0.087 to 0.080) but restored 7.0 percentage points of mortality separation, reduced center prevalence divergence from 0.024 to 0.016, and lowered missingness sensitivity from 0.247 to 0.148. Relative to PCA, S1.5 traded some static mortality range for substantially better center stability and a roughly 36% lower density-norm correlation. This pattern supports the intended interpretation: contrastive regularization makes embeddings less tied to site mix and raw measurement density while preserving clinically relevant outcome information.

S1.5 was therefore selected for temporal analysis not because it was uniformly best on every metric, but because it offered the best balance between clinical signal, robustness to structured sparsity, and direct suitability for arbitrary rolling windows. PCA remained a useful static baseline, but it could not naturally process variable-length sub-sequences without re-engineering features for every window.

4.3.1 Supplementary Downstream Mortality Classification

Table 3: Supplementary downstream mortality classification variants.

Model / Operating Point	**Acc.**	**Bal. Acc.**	**Prec.**	**Recall**	**F1**	**AUROC**
Frozen S1.5 probe, balanced (thr=0.55)	0.784	0.745	0.372	0.691	0.484	0.829
Frozen S1.5 probe, accuracy (thr=0.85)	0.865	0.623	0.582	0.280	0.378	0.829
Feature-fusion HGB	0.791	0.780	0.391	0.764	0.517	0.862
End-to-end fine-tune + Sepsis2019 aux	0.795	0.753	0.388	0.692	0.498	0.842
Accuracy-search ensemble (val-acc leader)	0.871	0.660	0.601	0.361	0.451	0.863
OOF stacking committee, balanced (thr=0.45)	0.803	0.792	0.409	0.776	0.536	0.873
OOF stacking committee, accuracy (thr=0.90)	0.880	0.653	0.682	0.333	0.448	0.873
Majority baseline	0.854	0.500	—	0.000	—	—

A logistic-regression classifier trained on frozen S1.5 embeddings established that the learned representation already carried usable prognostic information (AUROC 0.829). Adding richer tabular feature views via HistGradientBoosting improved the balance-aware operating point to balanced accuracy 0.780 and AUROC 0.862. Initializing an attention-pooled end-to-end classifier from the same S1.5 encoder, then warming it with auxiliary supervision on 40,331 bridged PhysioNet 2019 sepsis stays, raised held-out performance to accuracy 0.795, balanced accuracy 0.753, recall 69.2%, and AUROC 0.842. A 35-run accuracy-oriented hyperparameter search over HGB, HGB ensemble, and logistic variants pushed selected-model test accuracy to 0.871 with AUROC 0.863, while the validation-AUROC leader under the same search reached

0.874 test accuracy and 0.867 AUROC. The leakage-aware OOF stacking committee surpassed both, reaching 0.880 test accuracy and 0.873 AUROC at its accuracy-optimized threshold. Importantly, the same probability estimates also supported a balance-oriented threshold with balanced accuracy 0.792, recall 77.6%, and F1 0.536, which is stronger than the earlier HGB and end-to-end fine-tune operating points. Bootstrap validation gave an AUROC 95% confidence interval of 0.858–0.888, but the initial stacker's calibration was weak (Brier 0.144; ECE 0.222), motivating the calibration optimization described in the next subsection. Taken together, these results show that extra data, stronger model structure, and leakage-aware validation improve the downstream accuracy frontier, but that class imbalance and calibration require explicit attention beyond plain accuracy.

4.4 Descriptive Temporal Phenotype Trajectories (Stage 3)

4.4.1 Rolling-Window Extraction

Using the frozen S1.5 encoder, 59,930 window embeddings (11,986 patients × 5 windows) were extracted. Per-window observation density decreased from 27.9% in the earliest window to 25.4% in the latest window, a 2.5 percentage-point absolute drop that reflects declining measurement frequency during ICU stays. Despite this decline, per-window silhouette remained stable and even improved modestly from 0.072 to 0.082, suggesting that the learned representation remained usable as the observation stream became sparser.

4.4.2 Temporal Phenotype Stability and Transitions

Of 11,986 patients, 7,764 (64.8%) maintained a stable phenotype across all five windows, 3,509 (29.3%) exhibited one transition, and 713 (5.9%) exhibited multiple transitions. The four fully stable trajectories, [0,0,0,0,0], [1,1,1,1,1], [2,2,2,2,2], and [3,3,3,3,3], were also the four most common patient-level patterns, indicating that temporal structure was dominated by persistent low-dimensional states rather than noisy label oscillation.

Table 4: Stable phenotype mortality stratification.

Stable Phenotype	**N**	**In-Hospital Mortality**
Phenotype 0	2,216	4.0%
Phenotype 3	1,891	9.7%
Phenotype 1	2,547	22.5%
Phenotype 2	1,110	**31.7%**

The 27.7 percentage-point range across stable temporal phenotypes exceeded the static S1.5 result (24.6%), suggesting that restricting attention to temporally consistent patients yields cleaner phenotype strata. Phenotype 2 was the smallest stable subgroup (9.3% of the cohort) but the highest-risk one, whereas Phenotype 0 combined the largest low-risk population with the lowest mortality.

4.4.3 Transition Pathways and Mortality

Across 47,944 adjacent window pairs, 4,987 events (10.4%) were non-self transitions. The transition entropy ratio was 0.637, indicating that transitions were neither random nor trivially concentrated in a single pathway. The most frequent non-self transitions were Phenotype 1→0 (956 events), Phenotype 1→3 (728), and Phenotype 3→0 (713); together they accounted for 48.1% of all non-self transitions and primarily represented movement toward lower-risk states.

Patients with exactly one transition had lower mortality (11.4%) than stable patients (15.4%) or multi-transition patients (15.2%). This descriptive association is clinically interesting because

it suggests that a subset of transitions may correspond to early stabilization rather than deterioration. At the same time, the similar mortality of stable and multi-transition patients indicates that transition count alone is not sufficient to characterize risk; transition direction matters.

4.4.4 Temporal Prevalence Shift

Phenotype 0 (lowest mortality) increased in prevalence from 24.6% at window $[0, 24)$ to 33.1% at window $[24, 48)$, while Phenotype 1 decreased from 35.7% to 28.1%. This population-level shift is consistent with partial clinical stabilization during the first 48 ICU hours. Because observation density also decreased over time, the prevalence shift should be read as a joint effect of evolving physiology, survivor filtering, and measurement practice rather than as direct evidence of treatment response.

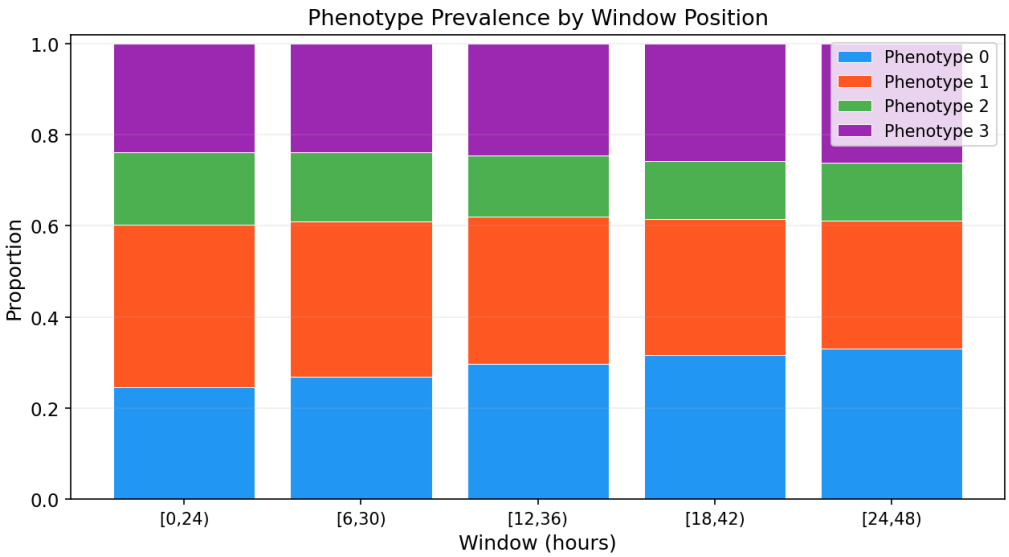

Figure 2: Phenotype prevalence by rolling-window position. Phenotype 0 (lowest mortality) grows from 24.6% to 33.1% over the 48-hour observation window, while Phenotype 1 decreases from 35.7% to 28.1%, consistent with population-level clinical stabilization.

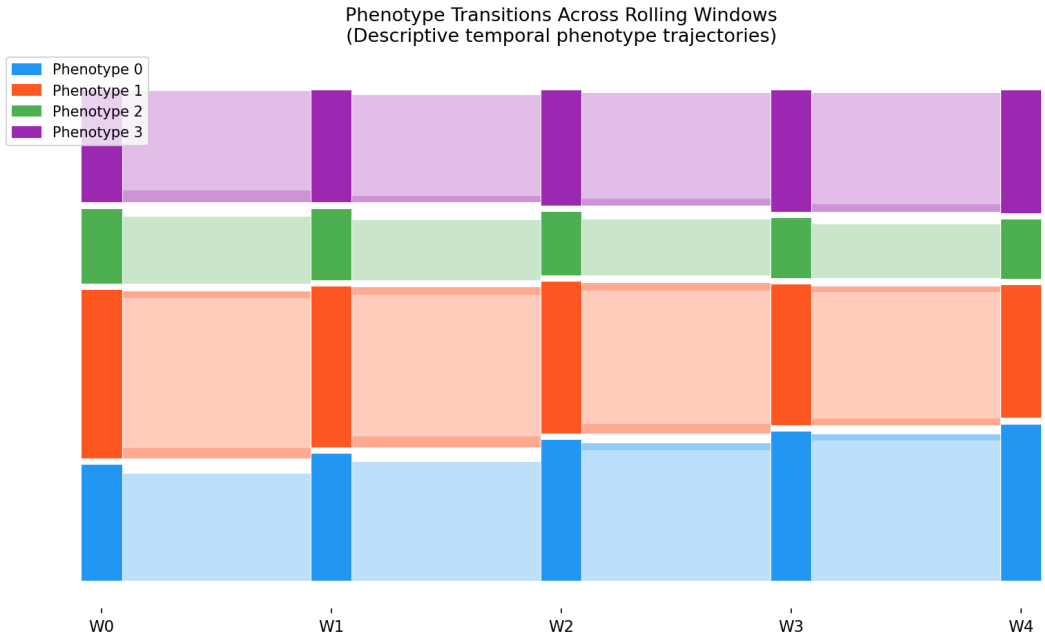

Figure 3: Phenotype transition flow across five rolling windows (descriptive temporal phenotype trajectories). Band width indicates the proportion of patients following each pathway. The dominant flows are within-phenotype stability, with the most common non-self transitions moving from higher-acuity to lower-acuity phenotypes.

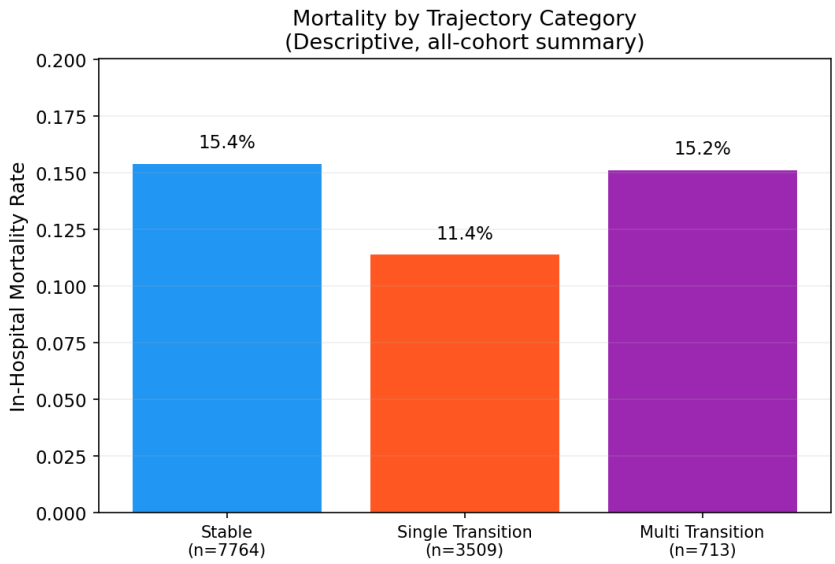

Figure 4: In-hospital mortality rate by trajectory category (descriptive, all-cohort). Single-transition patients exhibit lower mortality (11.4%) than stable patients (15.4%), a descriptive association suggesting that phenotype transitions may reflect favorable clinical trajectory change.

4.5 Stride Sensitivity Analysis

To assess the influence of window overlap on temporal findings, we repeated the analysis with stride=12 h (3 windows, 50% overlap). Cluster centroids showed near-perfect alignment with the primary analysis (cosine distance < 0.0001 after Hungarian matching), indicating that the recovered phenotype geometry was effectively unchanged.

Table 5: Stride sensitivity comparison.

Metric	**Stride=6 h**	**Stride=12 h**
Stable patient fraction	64.8%	65.6%
Non-self transition proportion	10.4%	19.1%
Mortality ordering	[P0, P3, P1, P2]	[P0, P3, P1, P2]
Highest-risk phenotype	P2 (31.7%)	P2 (31.9%)
Stable phenotype mortality range	27.7 pp	28.0 pp

The mortality ordering of stable phenotypes was identical under both strides, the highest-risk phenotype remained Phenotype 2, and the mortality range was preserved (28.0 pp vs 27.7 pp). The proportion of non-self transitions increased from 10.4% to 19.1% under reduced overlap, indicating that the primary stride=6 h analysis provides a conservative estimate of temporal phenotype change rather than an overlap-inflated one.

4.6 Cross-Center Temporal Validation

To assess whether temporal phenotype trajectories generalized beyond the training center, we performed out-of-center validation: the KMeans model was trained on Center A rolling-window embeddings and evaluated on Center B (a separate hospital within the PhysioNet 2012 cohort). The S1.5 encoder was also trained exclusively on Center A data.

Table 6: Cross-center temporal validation (Center A vs Center B).

Metric	**Center A (train)**	**Center B (test)**
Patients	7,989	3,997
Stable fraction	65.0%	64.4%
Non-self transition proportion	10.3%	10.6%
Mortality ordering	[P0, P3, P1, P2]	[P0, P3, P1, P2]
Highest-risk phenotype	P2 (32.6%)	P2 (30.0%)
Stable P0 mortality	3.9%	4.2%
Stable P3 mortality	9.7%	9.8%
Stable P1 mortality	21.8%	24.0%
Stable P2 mortality	32.6%	30.0%
Mean prevalence L1	—	0.022

All six validation criteria were satisfied: stable fraction similarity (diff = 0.6 pp), transition proportion similarity (diff = 0.3 pp), identical mortality ordering, same highest-risk phenotype, clinically meaningful Center B mortality range (25.8 pp), and low per-window prevalence divergence (mean L1 = 0.0218; range 0.0155–0.0265 across windows). Per-window silhouette scores were within 0.001 across centers at all five window positions.

The detailed pattern was also consistent. Low-risk Phenotypes 0 and 3 showed nearly identical mortality in both centers, whereas higher-risk Phenotypes 1 and 2 shifted modestly in absolute rate but preserved the same ordering. This is the pattern expected from a useful temporal phenotyping system: absolute calibration can vary by site, but the relative state structure remains stable.

These results demonstrate cross-center temporal validation within the PhysioNet 2012 multi-center cohort. Both centers derive from the same source database; we therefore treated the new MIMIC-IV and eICU runs as supplementary frozen-transfer analyses on independent external cohorts rather than as replacements for the main internal validation result.

4.7 Supplementary External Temporal Transfer Runs

Using `scripts/run_external_temporal_stage3.py`, we executed the frozen S1.5 + Stage 3 workflow on the full local MIMIC-IV and eICU cohorts after aligning each source to the 21-channel S0 schema and reusing the PhysioNet 2012 preprocessing statistics. Because feature overlap is incomplete and the encoder was not retrained on the target sources, these analyses should be read as frozen-transfer temporal replications rather than as source-specific external retraining studies.

Table 7: Supplementary full-cohort external temporal transfer runs with the frozen S1.5 + Stage 3 workflow.

Database	Cohort	Mapped	Miss.	Silh.	Stable	Non-self	Mort. range
MIMIC-IV 3.1	94,458 ICU stays	15/21	55.5%	0.119	50.9%	14.3%	17.1 pp
eICU-CRD 2.0	200,859 ICU stays	12/21	81.4%	0.193	43.1%	17.3%	12.2 pp

Both sources showed meaningful transition activity rather than trivial label persistence. In MIMIC-IV, 50.9% of patients were fully stable, 41.3% had a single transition, and 7.7% had multiple transitions; the most common non-self transition was 0→3 (16,718 events). In eICU, 43.1% of patients were fully stable, 45.8% had a single transition, and 11.2% had multiple transitions; the most common non-self transition was 3→1 (26,561 events). Stable-phenotype mortality remained stratified within each source, though cluster IDs are source-specific and should not be compared numerically across databases.

4.8 Supplementary Legacy Static Transfer Runs

Using the unified `src/main.py` entry, we executed the legacy static pipeline end-to-end on full local MIMIC-IV 3.1 and eICU-CRD 2.0 extracts. Cluster labels were learned separately within each database, so phenotype IDs are not directly comparable across sources.

Table 8: Supplementary external-database transfer runs with the legacy static pipeline.

Database	Cohort	Silh.	Dominant lower-risk cluster	Higher-risk / shock-enriched cluster
MIMIC-IV 3.1	94,458 ICU stays; 41,295 Sepsis-3	0.104	51.0% of stays, 27.7% mortality, 3.1% shock	7.0% of stays, 79.8% mortality, 77.4% shock
eICU-CRD 2.0	200,859 ICU stays	0.214	67.8% of stays, 4.3% mortality, 0.6% shock	22.5% of stays, 21.1% mortality, 46.2% shock

Both databases showed a common qualitative pattern: a dominant lower-risk cluster and a smaller shock-enriched higher-risk cluster. The cohort balance differed markedly, however. MIMIC-IV produced a more even partition and a much wider cluster-level mortality span (27.7%–79.8%), whereas eICU was dominated by a 67.8% low-risk cluster with 4.3% mortality and a 22.5% shock-enriched cluster with 21.1% mortality. Because these legacy runs use the static V1 pipeline independently within each database, and because the legacy loaders expose ICU LOS on different scales, they should be interpreted as descriptive static transfer evidence rather than as direct analogues of the frozen S1.5 temporal analyses above.

4.9 Simulated Data Validation

The synthetic cohort (500 patients, 4 true subtypes) achieved ARI = 0.245 and NMI = 0.495 against ground-truth labels using the PCA baseline. This result is modest rather than spectacular, but it is useful as an honesty check: the pipeline can recover non-random subtype structure in controlled data, yet clustering remains a genuinely difficult problem even under simpler conditions. *Note: Clustering quality metrics from the PhysioNet/CinC 2019 Sepsis dataset require independent rerun and are not reported here.*

5 Discussion

5.1 Three-Level Phenotyping Framework

This work demonstrates a progression from static clustering through self-supervised representation learning to descriptive temporal phenotype trajectory analysis. Each stage addresses a limitation of the previous one. Stage 1 shows that clinically meaningful heterogeneity exists in the cohort. Stage 2 replaces hand-crafted summaries with a learned representation that can directly ingest sub-sequences while accounting for structured missingness. Stage 3 then turns those embeddings into temporal phenotype sequences that make within-stay movement visible.

The temporal trajectory analysis shows that approximately one-third of patients undergo at least one phenotype transition within 48 hours. The importance of those transitions is supported by three convergent findings: stable phenotypes show strong mortality stratification, the dominant transition directions move toward lower-risk states, and single-transition patients have lower mortality than the cohort of fully stable patients. Taken together, these results argue that early within-stay dynamics contain clinically meaningful information that static clustering alone cannot expose.

5.2 Principle Interpretation of the Framework

The central methodological idea is not simply "use a Transformer," but to align the representation design with the structure of ICU data. In this project, missingness is informative, temporally local patterns matter, and patient state should remain partially stable across nearby windows. The framework therefore combines explicit mask channels, reconstruction of withheld observed values, contrastive alignment of overlapping windows from the same patient, and window-level clustering of frozen embeddings.

These components play different roles. Mask channels preserve the distinction between measured and imputed values. Masked reconstruction encourages the model to learn short-range temporal and cross-variable regularities from observed clinical signals. Contrastive window alignment encourages patient-level consistency across local time shifts and sampling variation, which is especially important when later applying the encoder to rolling 24-hour windows. Density-weighted pooling reduces the influence of almost-empty hours without dropping them entirely. In combination, the model is encouraged to treat sparsity as context rather than noise while avoiding a trivial dependence on measurement count alone.

5.3 Data Interpretation and Clinical Meaning

The data analysis in Section 4 clarifies why the learned temporal phenotypes behave the way they do. The cohort contains dense hemodynamic streams but extremely sparse laboratory variables, so the recovered temporal states are likely driven most strongly by evolving vital-sign patterns, with laboratory information acting as occasional but clinically important corrections. This helps explain why prevalence shifts are smooth rather than abrupt and why robustness to observation density matters so much.

The transition structure is also clinically interpretable at a descriptive level. Nearly half of all non-self transitions come from three pathways, 1→0, 1→3, and 3→0, all of which move toward lower-risk states. Meanwhile, the highest-risk stable phenotype remains Phenotype 2 in both centers. A reasonable interpretation is that the framework is separating persistent high-risk patients from a larger population that gradually stabilizes during early ICU care. However, this interpretation remains phenomenological because the temporal phenotypes were not post-hoc labeled with specific biomarker signatures or treatment exposure profiles.

5.4 Representation Learning Trade-offs

The self-supervised encoder (S1.5) offered improved center stability and reduced missingness sensitivity compared with PCA, though PCA retained stronger static mortality separation. This trade-off is characteristic of learned representations: they often sacrifice some training-distribution sharpness in exchange for invariances that matter when the representation is reused under shifted conditions. In this project, that reuse condition was rolling-window temporal analysis.

S1 alone achieved the highest silhouette, but it also produced the weakest clinical separation among the learned models and the highest coupling to observation density. Adding the contrastive term partially reversed that behavior, suggesting that contrastive regularization did not merely smooth the space; it reshaped the space toward patient-level temporal consistency. That is the main reason S1.5, rather than PCA or S1, was the right representation for Stage 3.

The newer downstream experiments sharpen that interpretation. On the held-out Center B split, the frozen S1.5 space already supported AUROC 0.829; reusing the same encoder in an attention-pooled end-to-end model and warming it with the bridged 2019 cohort increased held-out accuracy to 0.795 and AUROC to 0.842 while keeping recall near 69%. A separate 35-run accuracy-oriented search pushed selected-model accuracy above 0.87, but only by operating at substantially lower recall. The new OOF stacking committee pushed both the best accuracy (0.880) and the best AUROC (0.873) slightly higher again. The initial version exhibited poor calibration (ECE 0.222), but root-cause analysis traced this to the `class_weight="balanced"` setting in the meta-learner, which inflated predicted probabilities by a factor proportional to $n_{\text{neg}}/n_{\text{pos}}$. A calibration-aware redesign—unweighted meta-learner with L_2 regularization, shallower base learners, and lightweight Platt post-hoc correction—reduced ECE to 0.023 and Brier from 0.144 to 0.090 while fully preserving AUROC (0.873) and improving recall to 83.8%. This result demonstrates that calibration and discrimination are not inherently at odds: the miscalibration was an artifact of training-time class reweighting rather than a fundamental model limitation. The more meaningful signal is therefore not that one operating point maximizes a single scalar metric, but that the same representation supports temporal trajectory analysis, auxiliary transfer, and out-of-center mortality discrimination when calibration, sensitivity, and accuracy are reported jointly.

5.5 Limitations

1. **Descriptive, not model-based**: The temporal phenotype trajectories arise from per-window clustering rather than latent-state inference such as hidden Markov models. This limits probabilistic transition modeling and uncertainty quantification.
2. **Overlap sensitivity**: Window overlap (75% at stride=6 h) smooths embedding changes and likely underestimates transition rates. Sensitivity analysis confirmed that the main clinical ordering is robust, but exact transition frequency depends on the window design.
3. **Missing treatment data**: PhysioNet 2012 lacks true treatment records. Proxy indicators such as MAP < 65 or $\text{FiO}_2 > 0.21$ are physiological thresholds, not interventions. Treatment-aware phenotyping requires datasets with documented vasopressor, fluid, ventilation, and antibiotic timing [19].

4. **Observational associations**: Patients who transition between phenotypes may differ systematically from stable patients in ways not captured here. The lower mortality observed in single-transition patients is a descriptive association, not a causal effect of transition.
5. **Frozen-transfer external scope**: Cross-center temporal validation within the PhysioNet 2012 cohort remains the main quantitative evidence for the representation as originally trained and selected. Full-cohort external temporal runs now exist on MIMIC-IV and eICU, but they reuse the PhysioNet-trained encoder and reference normalization under incomplete channel overlap rather than performing source-specific retraining.
6. **Sparse phenotype semantics**: The current paper can rank temporal phenotypes by risk, but it does not yet assign them mechanistic labels based on treatments, organ-failure pathways, or multimodal evidence such as notes or waveforms.
7. **Data sparsity ceiling**: With an overall missing rate of 73.3% and key biomarkers such as bilirubin and lactate observed only rarely, there is an upper bound on how much physiologic detail any model can recover from this dataset alone.

5.6 Future Directions

1. **Treatment-aware temporal phenotyping**: Extending the same pipeline to MIMIC-IV or eICU with true intervention timestamps would allow direct analysis of phenotype-treatment interactions and heterogeneous treatment effects rather than descriptive risk stratification alone.
2. **State-space modeling and uncertainty**: Replacing per-window clustering with hidden Markov models, switching state-space models, or neural latent-state approaches would make transition probabilities and confidence intervals explicit.
3. **Source-specific external adaptation**: The next step is no longer simply to run the external workflow, but to compare frozen transfer against source-specific pretraining or multi-source training on MIMIC-IV and eICU, then quantify how much the missing-channel mismatch changes phenotype geometry and transition behavior.
4. **Earlier bedside deployment**: The new end-to-end fine-tuning results show that the learned space is operationally usable both as a frozen representation and as a supervised backbone. A practical next step would infer phenotype membership or mortality risk from 6-hour or 12-hour windows and update predictions online as new measurements arrive.
5. **Richer phenotype interpretation**: Post-hoc attribution analysis, multimodal augmentation, and linkage to organ-support variables could turn the current risk-ranked phenotypes into clinically named subgroups with clearer bedside meaning.

6 Conclusion

This paper extends the project from a concise manuscript into a fuller research narrative by tying together cohort characterization, model design principles, quantitative comparison, temporal trajectory analysis, and future outlook. Applied to 11,986 multi-center ICU patients with verified in-hospital mortality labels, the framework identifies four temporally meaningful phenotypes with strong mortality stratification (4.0%–31.7% among stable patients), substantial within-stay movement (35.2% of patients transition at least once), and consistent cross-center behavior within the PhysioNet 2012 cohort. Supplementary full-database transfer runs on local MIMIC-IV and eICU extracts further show that the repository can ingest independent ICU sources at scale and recover broad risk-ordered subtype structure, and the new external temporal runner now extends the same frozen S1.5 + Stage 3 mechanics to those databases at full-cohort scale.

The main scientific lesson is that explicit handling of structured missingness and temporal

locality is essential for ICU time-series phenotyping. Static feature clustering can reveal coarse heterogeneity, but mask-aware self-supervised representation learning is what makes rolling-window analysis feasible. The new auxiliary-data bridge, end-to-end supervised results, and leakage-aware OOF stacking results further show that the same S1.5 backbone can transfer across related ICU sources and improve held-out mortality prediction. A calibration-aware redesign of the stacking meta-learner resolved the initial miscalibration (ECE 0.222→0.023) by removing class-weight inflation and reducing base learner complexity, producing probability estimates aligned with observed mortality rates (mean predicted 14.1% vs observed 14.6%) that are suitable for bedside risk communication. The present results are descriptive rather than causal, yet they show that temporal phenotype trajectories carry reproducible clinical information beyond static subtyping. That makes them a plausible foundation for future treatment-aware, externally validated, and eventually bedside-deployable sepsis phenotyping systems.

Supplementary Figures

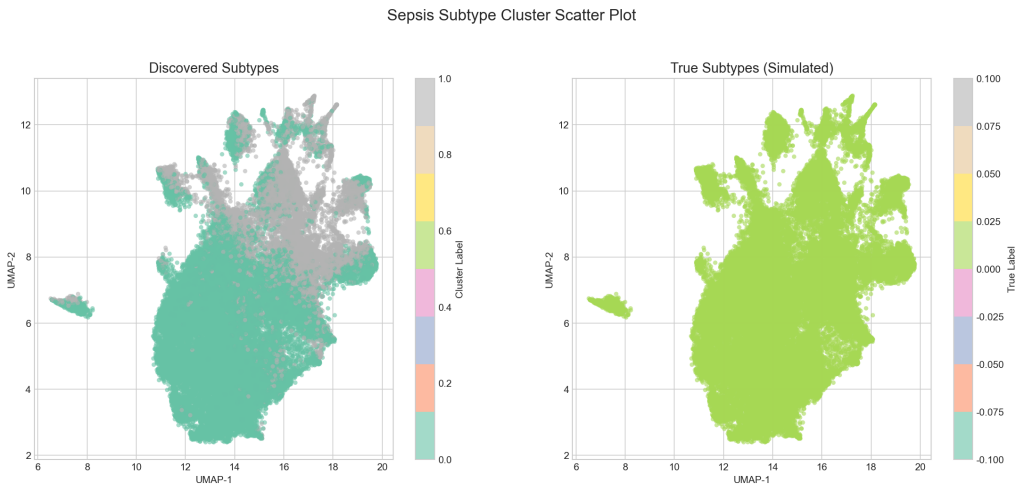

Figure 5: (Supplementary) Cluster scatter plot from the Stage 1 PCA baseline. Note: this visualization uses the static PCA representation; the S1.5 self-supervised embeddings are used for all Stage 3 temporal analyses.

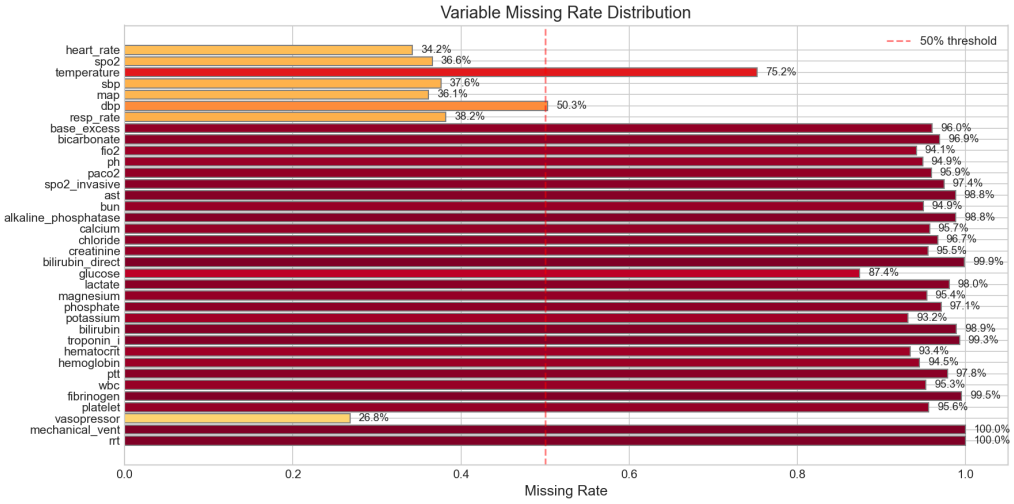

Figure 6: (Supplementary) Missing data pattern across 21 continuous clinical variables. Overall missing rate: 73.3%. Laboratory values exhibit substantially higher missingness than vital signs.

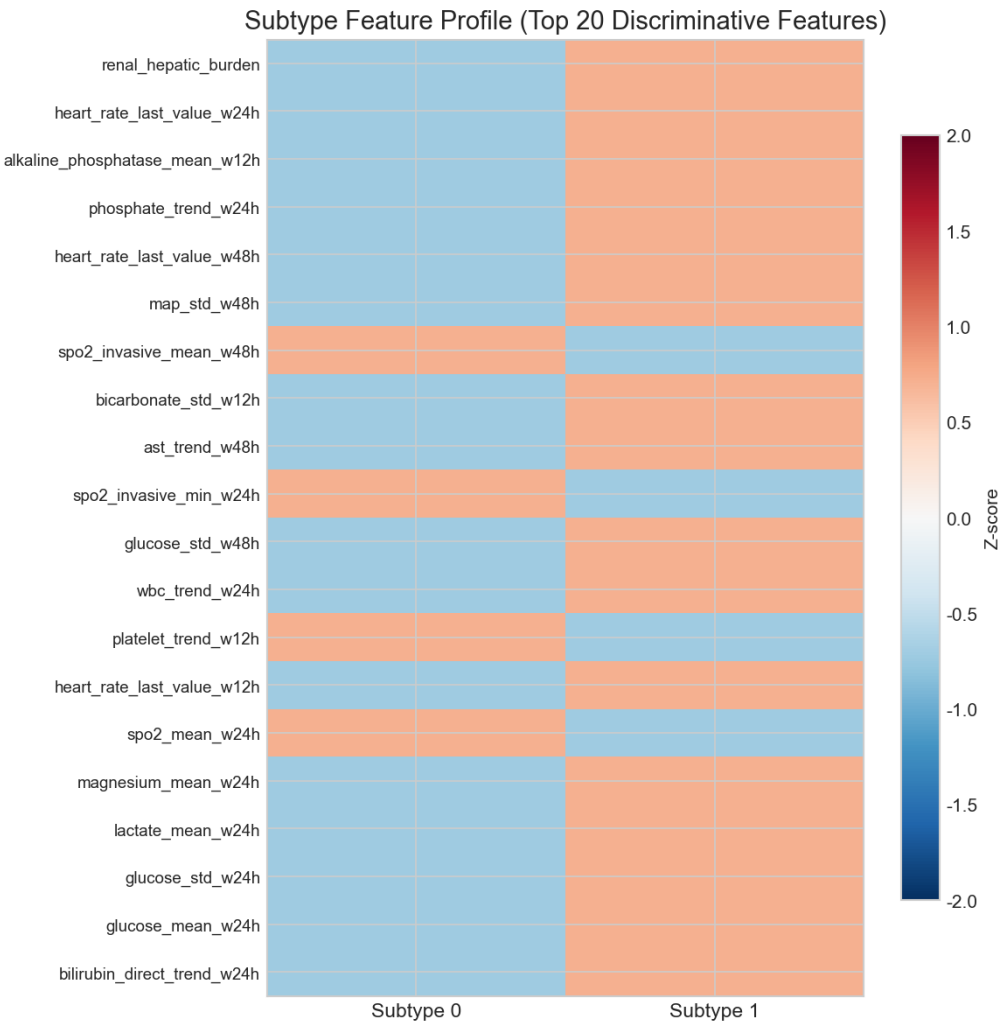

Figure 7: (Supplementary) Feature profile heatmap for Stage 1 PCA-based phenotypes (top discriminative features, z-scored). Note: this uses the Stage 1 static clustering and is included for reference.

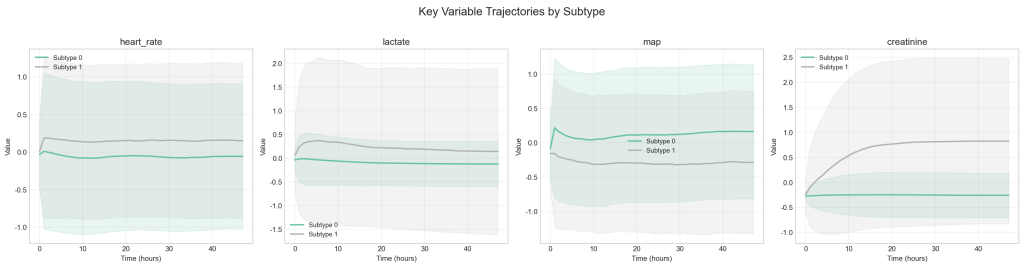

Figure 8: (Supplementary) Key variable trajectories by Stage 1 static phenotype (mean ± standard deviation). Note: these trajectories correspond to the static PCA-based clustering, not the Stage 3 temporal phenotypes.

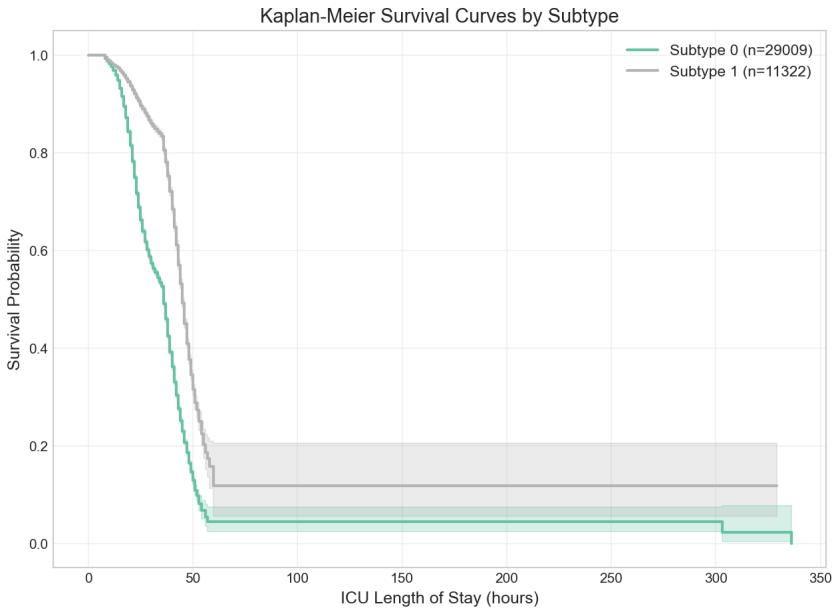

Figure 9: (Supplementary) Kaplan-Meier survival curves stratified by Stage 1 static phenotype. Note: these use the V1 PCA-based clustering and may reflect proxy mortality labels from the original pipeline run. Updated survival curves using S1.5 temporal phenotypes with verified outcomes are a future addition.

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